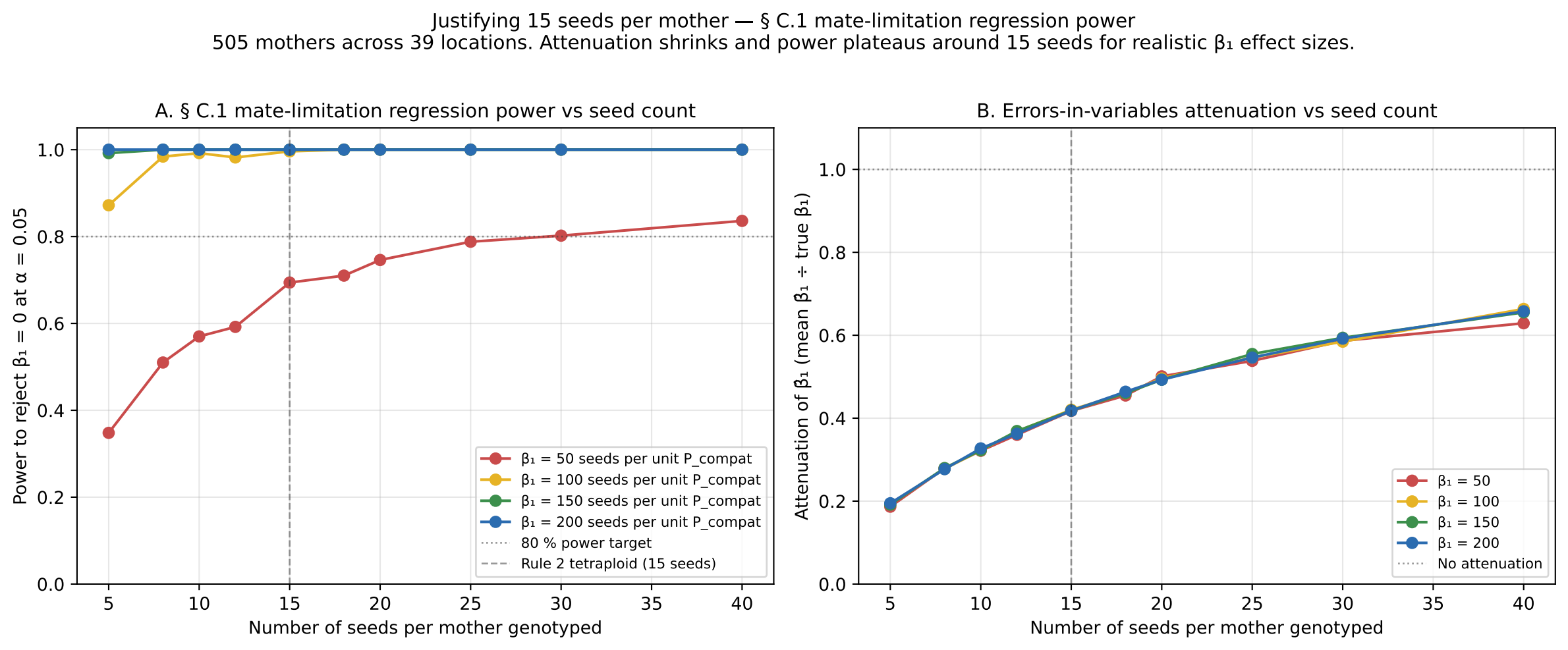
Justifying 15 seeds per mother — § C.1 mate-limitation regression power
505 mothers across 39 locations. Attenuation shrinks and power plateaus around 15 seeds for realistic β₁ effect sizes.
A. § C.1 mate-limitation regression power vs seed count
Power to reject β₁ = 0 at α = 0.05
Number of seeds per mother genotyped
β₁ = 50 seeds per unit P_compat
β₁ = 100 seeds per unit P_compat
β₁ = 150 seeds per unit P_compat
β₁ = 200 seeds per unit P_compat
80 % power target
Rule 2 tetraploid (15 seeds)
B. Errors-in-variables attenuation vs seed count
Attenuation of β₁ (mean β₁ ÷ true β₁)
Number of seeds per mother genotyped
β₁ = 50
β₁ = 100
β₁ = 150
β₁ = 200
No attenuation